RENORMALIZED TYPE II LIMITS AND REPAIRED GAUGES FOR NAVIER–STOKES BLOWUP

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Abstract. We formulate and prove the representation and PDE statements needed for a repaired-gauge analysis of Type II blowup solutions for the three-dimensional incompressible Navier–Stokes equations. Starting from explicit concentration, profile completeness, and gauge solvability hypotheses, we construct a renormalized Navier–Stokes orbit in a fixed scale and translation gauge, prove the pressure and modulation identities, replace exact critical normalization by a positive finite critical annulus, and identify the localized dissipation cost used by the compactness criterion. Under the stated Navier–Stokes compactness implication, every finite-cost Type II solution satisfying the hypotheses that is not excluded by the compactness criterion is either noncompact/radiative or has unbounded local windowed enstrophy in a bounded renormalized core.

1. Introduction

We consider possible finite-time Type II singular behavior for solutions of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

on a time interval ending at a putative singular time. This paper does not address global regularity; it isolates a precise set of standard analytic hypotheses under which a Type II solution can be placed in a renormalized repaired gauge and then analyzed by compactness alternatives.

The paper has four parts. First, we state the Type II hypotheses in standard PDE terms: concentration extraction modulo Navier–Stokes symmetries, profile completeness, and absence of Type II solutions outside the stated alternatives. Second, we prove that an absolutely continuous concentration chart and an absolutely continuous repaired-gauge selection produce the renormalized Navier–Stokes equation with the correct pressure and modulation coefficients. Third, we show that the compactness argument does not require exact normalization $\|V(\tau)\|_{L^3} = 1$; a uniform positive finite critical annulus suffices. Finally, we identify the localized Navier–Stokes cost used by the compactness criterion and obtain a trichotomy: compact Type II solutions are excluded by the compactness criterion, while non-excluded finite-cost solutions are either radiative or rough-core.

The hypotheses are stated explicitly below. The notation is the standard notation of rescaled Navier–Stokes analysis. No conclusion below asserts that all weak Navier–Stokes singularities satisfy these assumptions, nor does the paper address Type I solutions. The pressure is used in the standard sense, with the usual freedom to add functions of time; see, for example, [**?**, **?**, **?**, **?**] and the Calderon–Zygmund theory in [**?**, **?**]. Compactness statements are understood in the usual Aubin–Lions/Simon sense [**?**, **?**], and critical-space background may be found in [**?**, **?**].

2. Type II hypotheses

Definition 2.1 (Type II solution satisfying the hypotheses)**.** A Type II solution satisfying the hypotheses is a triple (u, p, T^*), with $T^* < \infty$, satisfying the Navier–Stokes equations on a terminal time interval and belonging to a prescribed class of Type II solutions. This means that the solution has not been identified as continuation, Type I, a boundary/open-system failure, or a non-Navier–Stokes domain failure.

Assumption 2.2 (Concentration and profile completeness)**.** Every Type II solution under consideration satisfies the following four properties.

(i) There is a concentration extraction modulo the Navier–Stokes scaling and translation symmetries.
(ii) The extracted solution lies in the supercritical Type II scaling regime used by the compactness criterion.
(iii) The extracted concentration object belongs to the Navier–Stokes profile class under consideration.
(iv) No Type II solution in the class under consideration escapes this concentration-scale-profile conclusion.

Theorem 2.3 (Concentration-profile conclusion for the class of Type II solutions satisfying the hypotheses)**.** *Under the concentration and profile completeness hypothesis, every Type II solution satisfying the hypotheses enters the concentration-scale-profile class described in the preceding assumption.*

Proof. Let (u, p, T^*) be a Type II solution satisfying the hypotheses. The concentration extraction property gives a concentration profile or blowup germ modulo the Navier–Stokes symmetry group. The scale classification property places this object in the supercritical Type II scaling regime. Profile completeness then places the extracted object in the Navier–Stokes profile class. The no-loss hypothesis excludes any additional Type II alternative outside these conclusions. Since the solution was arbitrary, the class of Type II solutions satisfying the hypotheses is exhausted by this classification. □

3. Renormalized charts and repaired gauges

3.1. Raw concentration chart.

Definition 3.1 (Raw concentration chart)**.** A raw concentration chart for a terminal solution consists of absolutely continuous functions

$$x_c : (t_0, T^*) \to \mathbb{R}^3, \qquad \lambda : (t_0, T^*) \to (0, \infty),$$

with $\lambda(t) \to 0$ as $t \uparrow T^*$, and a strictly increasing absolutely continuous time variable ρ satisfying

$$\frac{d\rho}{dt} = \lambda(t)^{-2}.$$

The raw variables (U, Q) are defined by

$$y = \frac{x - x_c(t)}{\lambda(t)}, \qquad u(x, t) = \lambda(t)^{-1} U(y, \rho(t)),$$

and

$$p(x, t) = \lambda(t)^{-2} Q(y, \rho(t)) + c(t),$$

where $c(t)$ is scalar. We assume enough local integrability to justify the distributional chain rule on compact sets in the y-variable.

Lemma 3.2 (Raw renormalized equation)**.** *Let (U, Q) be obtained from a raw concentration chart. Define*

$$A(\rho(t)) = \lambda(t)\lambda_t(t), \qquad B(\rho(t)) = \lambda(t)x_c'(t).$$

Then, distributionally on compact sets in the renormalized variable,

$$\partial_\rho U + (U \cdot \nabla)U + \nabla Q = \nu \Delta U + A(\rho)(U + y \cdot \nabla U) + B(\rho) \cdot \nabla U, \qquad \nabla \cdot U = 0.$$

Proof. The spatial derivatives transform as

$$\nabla_x u = \lambda^{-2}\nabla_y U, \qquad \Delta_x u = \lambda^{-3}\Delta_y U, \qquad (u\cdot\nabla_x)u = \lambda^{-3}(U\cdot\nabla_y)U.$$

Moreover

$$y_t = -\frac{x_c'}{\lambda} - \frac{\lambda_t}{\lambda}y, \qquad \rho_t = \lambda^{-2},$$

so

$$\partial_t u = \lambda^{-3}\partial_\rho U - \lambda^{-2}\lambda_t(U + y\cdot\nabla U) - \lambda^{-2}x_c'\cdot\nabla U.$$

Since $\nabla_x p = \lambda^{-3}\nabla_y Q$, substitution into the physical Navier–Stokes equation and multiplication by λ^3 gives the stated equation. The divergence equation follows from $\nabla_x\cdot u = \lambda^{-2}\nabla_y\cdot U$. □

3.2. **Repaired gauge.** Fix $0 < p < 3$, $\Theta_0 > 0$, a smooth cutoff ψ_R, and define

$$G_{\mathrm{sc}}(V) = \int_{\mathbb{R}^3} |Y|^{-p}|V(Y)|^3\,dY - \Theta_0,$$

$$G_j(V) = \int_{\mathbb{R}^3} Y_j|V(Y)|^2\psi_R(Y)\,dY, \qquad j = 1,2,3.$$

A repaired-gauge path is an absolutely continuous choice of $\mu(\tau) > 0$, $q(\tau)\in\mathbb{R}^3$, and $\rho = \rho(\tau)$, satisfying

$$\rho_\tau = \mu(\tau)^2,$$

such that

$$V(Y,\tau) = \mu(\tau)U(\mu(\tau)Y + q(\tau), \rho(\tau)),$$

$$P(Y,\tau) = \mu(\tau)^2 Q(\mu(\tau)Y + q(\tau), \rho(\tau)),$$

and

$$G_{\mathrm{sc}}(V(\tau)) = 0, \qquad G_j(V(\tau)) = 0, \quad j = 1,2,3.$$

The hypotheses include the usual admissibility, transversality, and nondegeneracy assumptions needed to select such a path absolutely continuously.

Lemma 3.3 (Final physical chart)**.** *Let (U,Q) be a raw concentration chart and let (μ,q,ρ) be a repaired-gauge path. Set*

$$\Lambda(\tau) = \lambda(t(\rho(\tau)))\mu(\tau), \qquad X(\tau) = x_c(t(\rho(\tau))) + \lambda(t(\rho(\tau)))q(\tau).$$

Then Λ and X are absolutely continuous on compact final time intervals, $\Lambda > 0$, and

$$\frac{d\tau}{dt} = \Lambda(\tau)^{-2}.$$

Furthermore, if $x = X(\tau) + \Lambda(\tau)Y$, then

$$u(x,t) = \Lambda(\tau)^{-1}V(Y,\tau), \qquad p(x,t) = \Lambda(\tau)^{-2}P(Y,\tau) + c(t).$$

Proof. The raw time ρ is strictly increasing since $\rho_t = \lambda^{-2} > 0$ a.e.; hence it has an absolutely continuous inverse on compact subintervals. The functions $\lambda(t(\rho(\tau)))$ and $x_c(t(\rho(\tau)))$ are absolutely continuous, and so are Λ and X. Since $d\tau/d\rho = \mu^{-2}$ and $d\rho/dt = \lambda^{-2}$, one obtains $d\tau/dt = (\lambda\mu)^{-2} = \Lambda^{-2}$. The formulae for u and p follow by substituting $y = q + \mu Y$ into the raw chart. □

Lemma 3.4 (Final modulation coefficients)**.** *Define*

$$a(\tau) = \partial_\tau \log \Lambda(\tau), \qquad b(\tau) = \Lambda(\tau)^{-1} X_\tau(\tau)$$

where derivatives are understood a.e. Then

$$a = \mu^2 A(\rho) + \frac{\mu_\tau}{\mu},$$

and

$$b = \mu B(\rho) + \mu A(\rho) q + \frac{q_\tau}{\mu}.$$

In particular $a, b \in L^1_{\mathrm{loc}}$ on finite renormalized time intervals.

Proof. The identity $\Lambda = \lambda(\rho(\tau))\mu(\tau)$ gives

$$\partial_\tau \log \Lambda = \rho_\tau \partial_\rho \log \lambda + \partial_\tau \log \mu = \mu^2 A + \frac{\mu_\tau}{\mu}.$$

Similarly,

$$X = x_c(\rho(\tau)) + \lambda(\rho(\tau)) q(\tau).$$

Using $\partial_\rho x_c = \lambda B$, $\partial_\rho \lambda = \lambda A$, and $\rho_\tau = \mu^2$, we obtain

$$X_\tau = \mu^2 \lambda B + \mu^2 \lambda A q + \lambda q_\tau.$$

Dividing by $\Lambda = \lambda\mu$ gives the formula for b. Local integrability follows from absolute continuity, positivity of μ, and local integrability of A, B. $\square$

Theorem 3.5 (Repaired-gauge representation)**.** *Assume that a Type II solution satisfying the hypotheses has a raw concentration chart and an absolutely continuous repaired-gauge path. Then the repaired variables satisfy*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0,$$

distributionally on compact sets. The pressure satisfies

$$-\Delta P = \partial_i \partial_j (V_i V_j)$$

modulo functions of τ, and the critical norm is invariant:

$$\|V(\tau)\|_{L^3_Y} = \|u(t(\tau))\|_{L^3_x}.$$

Proof. The equation follows by applying the raw renormalized equation to $V(Y, \tau) = \mu U(\mu Y + q, \rho)$. The derivative identities are

$$\nabla_Y V = \mu^2 \nabla_z U, \qquad \Delta_Y V = \mu^3 \Delta_z U, \qquad (V \cdot \nabla_Y)V = \mu^3 (U \cdot \nabla_z)U,$$

with $z = \mu Y + q$, together with $\rho_\tau = \mu^2$. The coefficients are those of the preceding lemma. Incompressibility follows from $\nabla_Y \cdot V = \mu^2 \nabla_z \cdot U$.

Taking divergence of the physical Navier–Stokes equation gives $-\Delta p = \partial_i \partial_j (u_i u_j)$, modulo functions of time. Pulling this identity through the raw chart and then through the repaired gauge gives the displayed pressure equation for P. Finally,

$$u(X + \Lambda Y, t) = \Lambda^{-1} V(Y, \tau), \qquad dx = \Lambda^3 \, dY,$$

so $\|u(t)\|_{L^3_x} = \|V(\tau)\|_{L^3_Y}$. $\square$

4. Critical mass and nonzero normalization

Let

$$N_3(\tau) = \|V(\tau)\|_{L^3(\mathbb{R}^3)}.$$

The compactness criterion is often stated with exact normalization $N_3(\tau) = 1$. For Navier–Stokes one cannot impose such a condition by a time-dependent amplitude renormalization without changing the equation. The correct invariant substitute is a positive finite critical annulus:

$$0 < \eta \le N_3(\tau) \le M < \infty \qquad (\tau \ge \tau_0).$$

Theorem 4.1 (Nonzero-mass compactness criterion)**.** *Let (V, P, a, b) be a repaired-gauge Type II solution. Assume:*

(i) $0 < \eta \le \|V(\tau)\|_{L^3} \le M < \infty$ *for all* $\tau \ge \tau_0$*;*

(ii) V *is uniformly tight in the critical norm, namely for every* $\varepsilon > 0$ *there is* R *such that*

$$\sup_{\tau \ge \tau_0} \int_{|Y|>R} |V(Y,\tau)|^3 \, dY < \varepsilon;$$

(iii) *for every* $m \ge 1$*,*

$$\sup_{n\in\mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|^2_{H^1(B_m)} \, d\tau < \infty.$$

Then the localized renormalization cost

$$\tilde{D}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(Y,\tau)|^2 \phi_{R_0}(Y) \, dY + a_+(\tau) \int_{\mathbb{R}^3} |V(Y,\tau)|^2 \phi_{R_0}(Y) \, dY$$

has infinite integral:

$$\int_{\tau_0}^{\infty} \tilde{D}_{R_0}(\tau) \, d\tau = \infty.$$

Proof. The standard good-window argument uses exact normalization only to preclude the zero limit. Suppose instead that the cost integral were finite. The compactness argument gives times $\sigma_j \to \infty$ such that, after passing to a subsequence, $V(\sigma_j) \to V_\infty$ strongly in L^3_{loc}, the local dissipation vanishes on the chosen exhaustion, and $\nabla V_\infty = 0$ distributionally on every fixed ball. Thus V_∞ is a constant vector. The upper bound $\|V(\sigma_j)\|_{L^3} \le M$ forces this constant to be zero by letting the radius of the ball tend to infinity.

Uniform critical tightness upgrades local convergence to global convergence to zero in L^3. Choose R so that the tail outside B_R is less than $\eta^3/4$, uniformly in τ. For large j, strong convergence on B_R gives an integral less than $\eta^3/4$ on B_R. Hence $\|V(\sigma_j)\|_{L^3} < \eta$, contradicting the assumed lower bound. □

Lemma 4.2 (Critical mass dichotomy)**.** *For a represented renormalized solution, after discarding a finite initial renormalized time interval if necessary, exactly one of the following ordered alternatives holds:*

(i) *the critical norm is not a well-defined extended measurable function;*

(ii) N_3 *is infinite somewhere or unbounded on the tail;*

(iii) N_3 *is bounded but* $\liminf_{\tau\to\infty} N_3(\tau) = 0$*;*

(iv) *there are* $0 < \eta \le M < \infty$ *such that* $\eta \le N_3(\tau) \le M$ *on the tail.*

Proof. If the norm is not well defined, the first alternative holds. Otherwise, if it is infinite somewhere or unbounded on the tail, the second alternative holds. If the supremum is finite but the liminf is zero, the third alternative holds. In the remaining case the liminf is positive and the supremum is finite; after moving τ_0 forward, the fourth alternative holds. □

5. Zero critical mass and terminal strata

Definition 5.1 (Nontrivial terminal core)**.** A represented terminal core is nontrivial on the tail if there are $R_* < \infty$, $\eta_* > 0$, and $\tau_* \geq \tau_0$ such that

$$\|V(\tau)\|_{L^3(B_{R_*})} \geq \eta_* \qquad (\tau \geq \tau_*).$$

Lemma 5.2 (Zero critical mass is incompatible with a nontrivial core)**.** *If a represented terminal core is nontrivial on the tail, then*

$$\liminf_{\tau \to \infty} \|V(\tau)\|_{L^3(\mathbb{R}^3)} > 0.$$

In particular the zero-critical-mass alternative in the preceding dichotomy is impossible.

Proof. The lower bound on B_{R_*} immediately implies

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} \geq \eta_* \qquad (\tau \geq \tau_*),$$

and hence the asserted positive liminf. □

Definition 5.3 (Terminal strata)**.** Assume that terminal profiles are partitioned into disjoint classes

$$\mathcal{S}_{\mathrm{core}}, \quad \mathcal{S}_{\mathrm{scatt}}, \quad \mathcal{S}_{\mathrm{ext}}, \quad \mathcal{S}_{\mathrm{rad}}, \quad \mathcal{S}_{\mathrm{rough}}, \quad \mathcal{S}_{\mathrm{multi}}.$$

The active classes are the core and multibubble classes after scattering, exterior, radiative, and rough-core alternatives have been removed. Each active terminal stratum $\mathfrak{s}$ carries a divergence-free profile $\Phi_{\mathfrak{s}} \in L^3(\mathbb{R}^3)$, and its critical mass is

$$N_{\mathfrak{s}} = \|\Phi_{\mathfrak{s}}\|_{L^3(\mathbb{R}^3)}.$$

Lemma 5.4 (Finite critical mass on active strata)**.** *Assume terminal profile completeness in the critical space and the usual small-data cutoff for inactive scattering profiles. Then every active terminal stratum satisfies*

$$0 < N_{\mathfrak{s}} < \infty.$$

If the active threshold is $\varepsilon_{\mathrm{sd}} > 0$, then $N_{\mathfrak{s}} \geq \varepsilon_{\mathrm{sd}}$.

Proof. An active stratum is represented by a nonzero terminal profile or finite compound profile after all scattering and exterior pieces have been removed. Thus its critical norm cannot vanish. Profile completeness gives membership in L^3, and finite sums of comparable-scale L^3 profiles remain in L^3. If the active list is defined by the small-data cutoff, being active means that the critical norm is at least the cutoff. □

Lemma 5.5 (Finite number of active strata)**.** *Let a terminal decomposition arise from a bounded critical sequence with*

$$\sup_n \|u_{0,n}\|_{L^3} \leq M_0.$$

If every active stratum has critical mass at least $\varepsilon_{\mathrm{sd}} > 0$, then the number of active strata is finite and at most

$$\left\lfloor \frac{M_0^3}{\varepsilon_{\mathrm{sd}}^3} \right\rfloor.$$

Moreover $N_{\mathfrak{s}} \leq M_0$ for each active stratum.

Proof. Critical mass decoupling gives, for every finite active subset A,

$$\sum_{\mathfrak{s} \in A} N_{\mathfrak{s}}^3 \leq M_0^3.$$

Since each active term is at least $\varepsilon_{\mathrm{sd}}^3$, the stated cardinality bound follows. Taking A to be a singleton gives the upper bound for each active stratum. □

6. Cost identification

The compactness theorem uses the localized nonnegative cost

$$\tilde{D}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(Y,\tau)|^2 \phi_{R_0}(Y)\, dY + a_+(\tau) \int_{\mathbb{R}^3} |V(Y,\tau)|^2 \phi_{R_0}(Y)\, dY.$$

We take this concrete Navier–Stokes quantity as the Type II cost in the compactness criterion.

Lemma 6.1 (Identity cost comparison)**.** *Assume $\tilde{D}_{R_0}$ is measurable, nonnegative, and locally integrable on finite renormalized time intervals. If*

$$\int_{\tau_0}^{\infty} \tilde{D}_{R_0}(\tau)\, d\tau = \infty,$$

then the cost-divergence condition in the Type II compactness criterion holds.

Proof. The comparison cost is chosen to be $\tilde{D}_{R_0}$ itself. Thus the identity comparison has constant one and no tail loss. Divergence of the Navier–Stokes localized cost is precisely the divergence condition used by the compactness criterion. □

Theorem 6.2 (Compactness criterion after cost identification)**.** *Assume that a repaired-gauge Type II solution satisfies the hypotheses of the nonzero-mass compactness criterion and that the localized cost above is measurable, nonnegative, and locally integrable on finite intervals. Then the intermediate compactness conclusion of the Type II compactness criterion holds. If, in addition, the Navier–Stokes compactness implication applies, then the compact renormalized Type II alternative is excluded.*

Proof. The nonzero-mass compactness criterion gives infinite localized cost. The identity cost comparison converts this into the divergence condition required by the Type II compactness criterion. The final exclusion conclusion is the separate Navier–Stokes compactness implication assumed in the statement. □

7. Radiative and rough-core alternatives

After the representation, critical-mass, and cost steps, the compactness criterion has only two analytic hypotheses left to check.

Definition 7.1 (Radiative and rough-core alternatives)**.** A represented renormalized solution is *radiative* if it fails uniform critical tightness, i.e. if there is $\varepsilon_0 > 0$ such that for every $R > 0$ there is $\tau_R \geq \tau_0$ with

$$\int_{|Y|>R} |V(Y,\tau_R)|^3\, dY \geq \varepsilon_0.$$

It is *rough-core* if it is critically tight but fails uniform local windowed H^1 control, i.e. if there is $m \geq 1$ such that

$$\sup_{n\in\mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|^2_{H^1(B_m)}\, d\tau = \infty.$$

Theorem 7.2 (Finite-cost Type II decomposition)**.** *Assume the Type II hypotheses, the repaired-gauge representation hypotheses, the positive finite critical annulus, the identity cost identification, and the Navier–Stokes compactness implication. Then every Type II solution satisfying the hypotheses satisfies one of the following alternatives:*

(i) *the compact renormalized Type II alternative is excluded by the compactness criterion;*
(ii) *the solution is radiative;*
(iii) *the solution is rough-core.*

Consequently, every finite-cost Type II solution satisfying the hypotheses that is not excluded by the compactness criterion is either radiative or rough-core.

Proof. Let a Type II solution satisfying the hypotheses be fixed. The concentration and profile hypotheses place it in the concentration-scale-profile class. The representation theorem gives a repaired-gauge orbit (V, P, a, b). The critical-mass result gives a positive finite critical annulus on the renormalized tail, and the cost identification places the localized Navier–Stokes cost in the compactness criterion.

Test uniform critical tightness. If it fails, the solution is radiative. If it holds, test uniform local windowed H^1 control. If that fails, the solution is rough-core. If both compactness hypotheses hold, the nonzero-mass compactness criterion gives infinite localized cost, the identity cost comparison gives the intermediate compactness condition, and the Navier–Stokes compactness implication excludes the compact renormalized Type II alternative. These alternatives are exhaustive by construction and disjoint in the stated order. □

Remark 7.3 (Scope of the decomposition)*.* The theorem is a Type II theorem under the explicit hypotheses of this paper. It does not exclude Type I behavior and does not assert that every weak Navier–Stokes singularity belongs to the class of Type II solutions satisfying the hypotheses. Its conclusion is that, within that class, finite-cost non-excluded solutions have only the two standard remaining mechanisms: radiative escape of critical mass or persistent roughness in a bounded renormalized core.

References